MULTITASK ATTENTIVE NETWORK FOR TEXT EFFECTS QUALITY ASSESSMENT (SUPPLEMENTARY)

Anonymous ICME submission

1. IMPLEMENTATION DETAILS

1.1. Network Architecture

The detailed architecture of the proposed multitask autoencoder network and the whole prediction model is shown in Fig. 1.

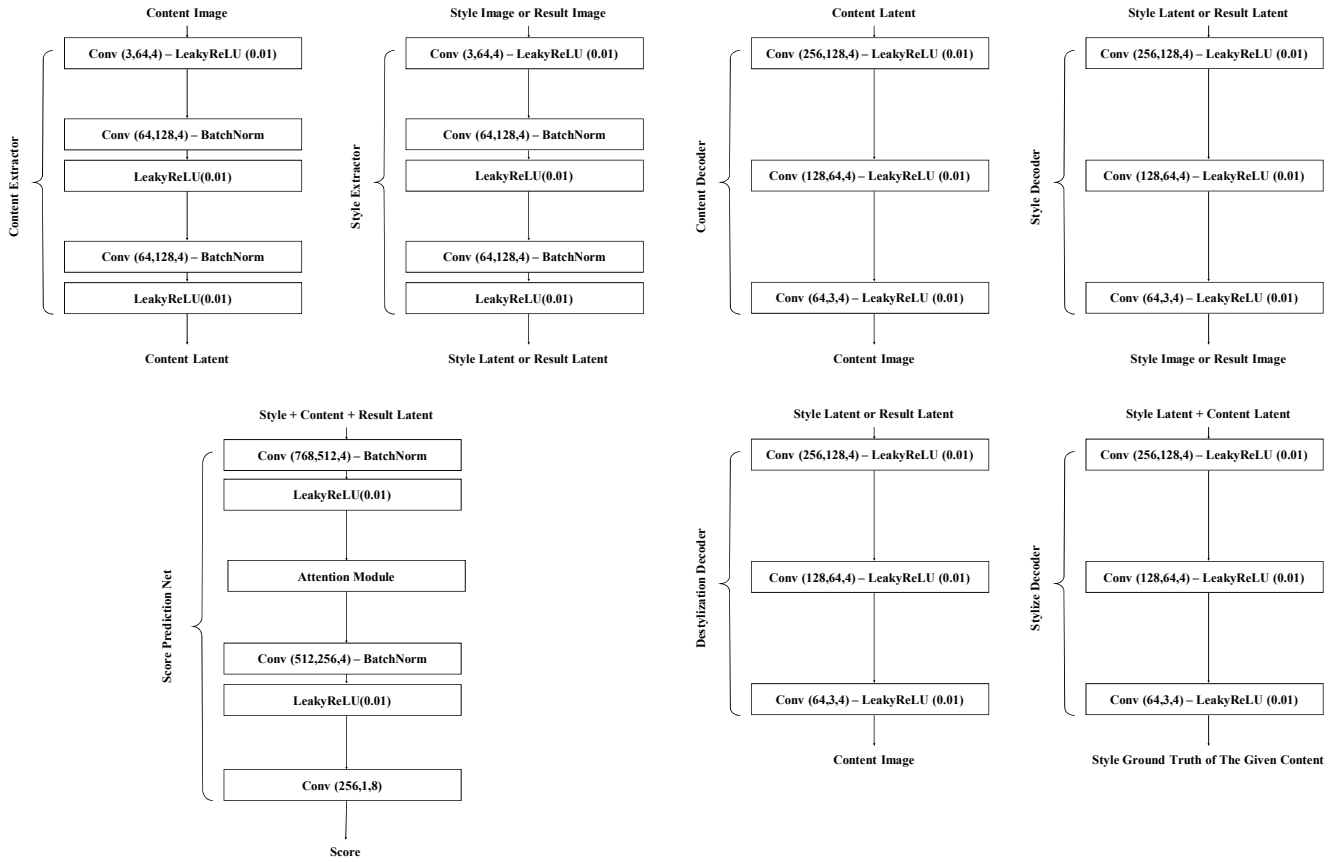

Fig. 1: Detailed architecture of the proposed multitask attention neural network.

1.2. Network Training

For the training data of multitask autoencoder, we randomly select a style, a random content reference and its corresponding golden like style ground truth from TE141K-S subset, then we randomly choose one result image of a style transfer method from the given 16 methods to form a pair of training data. The network is trained with content reconstruction loss $\lambda_{\mathcal{C}}^{AE}$, stylization loss λ^{sty}, destylization loss λ^{des} and style reconstruction loss $\lambda_{\mathcal{S}}^{AE}$ for 50 epochs. The loss weights are set to: $\lambda_{\mathcal{C}}^{AE} = 0.25$, $\lambda^{sty} = \lambda_{\mathcal{S}}^{AE} = 1$, and $\lambda^{des} = 1.25$. We use a fixed batch size of 2. For optimization, we use Adam (Kingma and Ba 2014) with a fixed learning rate of 0.0001.

For the training data of the whole model, we use 668 images to train and 167 images to evaluate. We use the pre-trained content and style extractors for initialization, and then train the whole network using L_2 loss for 500 epoches. We use a fixed batch size of 1. For the training images, we first resize them to 300 and then random crop to 256. We also use random horizontal flip and random rotation. For optimization, we use Adam with a fixed learning rate of 0.00005.

1.3. TE141K [1] Style Selection

The styles we select in the TE141K dataset can be seen in Fig.2. All these styles are of high complexity and difficult to assess.

Fig. 2: Image samples of the 20 selected styles.

2. METHODS MOS IN TE141K

The methods we choose to evaluate in TE141K is shown in Table 1, and the corresponding MOS statistics of these methods are shown in Table 2. The MOS score is of high consistency of the visual quality of these selected text effect style transfer methods. Readers may refer to TE141K [1] for a more detailed introduction of each method, and may refer to Neural Style Transfer: A Review [2] for a comprehensive review on style transfer.

Table 1: Style transfer models evaluated in the experiment

Algorithm	Settings		
	Full	Supervised one-shot	Unsupervised one-shot
NST [3]			✓
AdaIN [4]			✓
WCT [5]			✓
Analogy [6]		✓	
Quilting [7]			✓
CNNMRF [8]			✓
Doodles [9]		✓	
T-Effect [10]		✓	
UT-Effect [11]			✓
Pix2pix [12]	✓	✓	
BicycleGAN [13]	✓		
StarGAN [14]	✓		
TET-GAN [15]	✓	✓	✓

Table 2: Statistic of MOS of selected style transfer models.

ID	Model	Mean opinion score		
		Average	Variance (content)	Variance (style)
#1	NST [3]	2.129	0.196	0.063
#2	AdaIN [4]	2.241	0.035	0.021
#3	WCT [5]	2.426	0.107	0.049
#4	Analogy [6]	2.141	0.084	0.024
#5	Quilting [7]	1.967	0.112	0.036
#6	CNNMRF [8]	2.156	0.082	0.027
#7	Doodles [9]	3.726	0.152	0.023
#8	T-Effect [10]	4.363	0.067	0.023
#9	UT-Effect [11]	2.689	0.134	0.050
#10	Pix2pix-F [12]	3.334	0.167	0.054
#11	Pix2pix-S [12]	2.892	0.052	0.023
#12	BicycleGAN [13]	3.070	0.095	0.069
#13	StarGAN [14]	2.439	0.079	0.041
#14	TET-GAN-F [15]	4.190	0.059	0.043
#15	TET-GAN-S [15]	4.303	0.033	0.012
#16	TET-GAN-U [15]	4.120	0.043	0.025
	Average	3.012	0.094	0.037

3. MORE RESULTS

More score prediction results of our model are illustrated in Fig.3. Our network can well model the process of human visual judgement and evaluate the matching of content and style simutaneously.

Fig. 3: More results of text effect assessment from the test set. For each pair, the input style and content reference are on the left, and the query input image is on the right. Our predicted score is the number on the left in each image while the ground truth MOS is shown in brackets.

4. REFERENCES

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